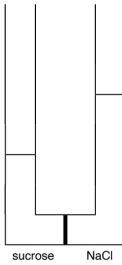
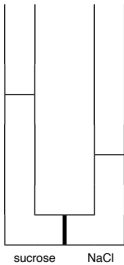
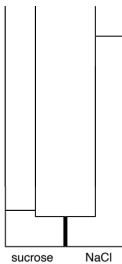
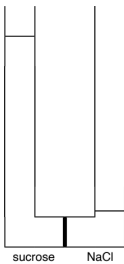


Stoichiometry/Solutions

- 10.00 g of blue copper sulfate pentahydrate, $\text{CuSO}_4 \cdot 5 \text{H}_2\text{O}$, is heated strongly to drive off all the water of hydration. How much colorless anhydrous copper sulfate remains?
(A) 1.67 g (B) 3.61 g (C) 6.39 g (D) 8.99 g
- Ethane reacts with oxygen according to the balanced equation below.
$$2 \text{C}_2\text{H}_6(\text{g}) + 7 \text{O}_2(\text{g}) \rightarrow 4 \text{CO}_2(\text{g}) + 6 \text{H}_2\text{O}(\text{l})$$

A mixture of ethane and oxygen occupies 22.4 L at STP. The mixture is allowed to react as described above, and the gases then occupy 11.2 L at STP. How many moles of ethane were originally present?
(A) 0.20 mol
(B) 0.30 mol
(C) 0.40 mol
(D) It cannot be determined from the information given.
- 0.2700 g of a white solid is titrated with 32.70 mL of 0.1834 M NaOH solution to achieve a phenolphthalein endpoint. Which is the molecular formula of the compound?
(A) CHO_2 (B) CH_2O_2
(C) $\text{C}_2\text{H}_2\text{O}_4$ (D) $\text{C}_2\text{H}_4\text{O}_4$
- Aqueous sodium hydroxide solutions absorb carbon dioxide both because carbon dioxide has an appreciable solubility in water and because the hydroxide ions react very favorably with excess CO_2 to form bicarbonate ions. A rigid vessel with 10.00 L of headspace contains 1.00 L of 0.100 M NaOH and is maintained at 25 °C. Carbon dioxide is introduced into the headspace at a pressure of 1.33 atm, which decreases to 1.00 atm after absorption into the aqueous NaOH is complete. If the experiment is repeated with an initial CO_2 pressure of 1.50 atm, 1.00 L of what concentration of NaOH would be required to reduce the pressure of CO_2 to 1.00 atm?
(A) 0.112 M (B) 0.152 M
(C) 0.169 M (D) 0.204 M
- What is the concentration of chloride ion in a solution prepared by mixing 0.100 L of 0.100 M LiCl with 0.200 L of 0.200 M BaCl_2 and 0.300 L of 0.300 M CeCl_3 ?
(A) 0.200 M (B) 0.233 M
(C) 0.467 M (D) 0.600 M
- A U-tube is divided into two compartments by a semipermeable membrane that allows only water to pass through. On the left side of the U-tube is added 10 mL of 0.20 M sucrose solution and on the right side is added 10 mL of 0.15 M sodium chloride solution. Which best represents the appearance of the U-tube after equilibrium is attained?
(A)  (B) 
(C)  (D) 
- Adipic acid, $\text{C}_6\text{H}_{10}\text{O}_4$ ($M = 146.15$), is prepared by the partial oxidation of cyclohexane, C_6H_{12} ($M = 84.16$), according to the following balanced reaction:
$$2 \text{C}_6\text{H}_{12}(\text{l}) + 5 \text{O}_2(\text{g}) \rightarrow 2 \text{C}_6\text{H}_{10}\text{O}_4(\text{s}) + 2 \text{H}_2\text{O}(\text{l})$$

A sample of 25.0 g cyclohexane reacts with excess oxygen to give 33.5 g of adipic acid. What is the yield of the reaction?
(A) 38.6% (B) 39.9% (C) 48.9% (D) 77.2%
- When a 1.000 g sample of CoCO_3 is heated under vacuum, it forms 0.630 g of an oxide of cobalt. When this sample is exposed to air, it forms a second oxide of cobalt, and attains a mass of 0.675 g. What are the formulas of the first and second oxide of cobalt, respectively?
(A) CoO , Co_2O_3 (B) CoO , Co_3O_4
(C) Co_2O_3 , Co_3O_4 (D) Co_3O_4 , Co_2O_3
- A white, crystalline solid contains only the elements C, H, N, and O. Complete combustion of a 1.000 g sample gives 1.831 g CO_2 ($M = 44.01$) and 0.750 g H_2O ($M = 18.02$). What is its molecular formula?
(A) $\text{C}_3\text{H}_6\text{NO}$ (B) $\text{C}_4\text{H}_4\text{NO}_2$
(C) $\text{C}_6\text{H}_{12}\text{N}_2\text{O}_2$ (D) $\text{C}_8\text{H}_8\text{N}_2\text{O}_4$

Stoichiometry/Solutions

10. 10.0 mL of 0.100 M CaCl_2 is mixed with 10.0 mL of 0.050 M AgNO_3 . What is the concentration of chloride ion in the final solution?
- (A) 0.025 M (B) 0.075 M
(C) 0.100 M (D) 0.200 M
11. A metal sulfate decahydrate contains 31.48% water by mass. What is the metal?
- (A) Na (B) Cr (C) Rh (D) Au
12. Which procedures will result in a 1.000 *m* solution of sodium acetate?
- I. Adding 1.000 mol sodium acetate trihydrate to a 1.000-liter volumetric flask, adding deionized water to dissolve, and then filling to the mark
II. Mixing 1.000 mol sodium hydroxide with 1.000 mol acetic acid and 1.000 kg deionized water
- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II
13. What is the mole fraction of acetic acid (CH_3COOH , $M = 60.05$) in an aqueous solution that is 11.7% by mass acetic acid?
- (A) 0.00195 (B) 0.0382
(C) 0.0398 (D) 0.195
14. Hydrogen peroxide ($M = 34.02$) decomposes to give water and oxygen gas. How much 3.00% by mass aqueous H_2O_2 must decompose to afford 4.00 L of dry O_2 at STP?
- (A) 6.08 g (B) 101 g (C) 203 g (D) 405 g
15. A diamagnetic compound that contains only carbon, hydrogen, nitrogen, and oxygen is 19.99% C, 3.35% H, and 23.31% N by mass. Which is its molecular formula?
- (A) $\text{CH}_2\text{N}_2\text{O}$ (B) CH_2NO_2
(C) $\text{C}_2\text{H}_3\text{NO}$ (D) $\text{C}_2\text{H}_4\text{N}_2\text{O}_4$
16. What is the chloride ion concentration in a solution prepared by mixing 35.0 mL of 0.35 M sodium chloride and 65.0 mL of 0.65 M calcium chloride?
- (A) 0.50 M (B) 0.54 M
(C) 0.85 M (D) 0.97 M
17. 17.0 g $\text{Ba}(\text{NO}_3)_2$ ($M = 261.32$) is mixed with 11.5 g of an alkali metal sulfate and the precipitated BaSO_4 ($M = 233.37$) is collected by filtration, placed in a tared crucible, and the crucible heated to drive off water. The mass of BaSO_4 obtained is 15.4 g. Which conclusion is best supported by the data?
- (A) The sulfate salt used was Na_2SO_4 ($M = 142.05$).
(B) The sulfate salt used was K_2SO_4 ($M = 174.27$).
(C) The BaSO_4 was not heated long enough to drive off all the water.
(D) Some of the BaSO_4 was spilled before it was transferred to the crucible.
18. 10.0 mL of 0.50 M potassium sulfate and 10.0 mL of 0.50 M silver nitrate solutions are mixed and the mixture is allowed to attain equilibrium. Which ordering of the concentrations of the ions in solution is correct?
- (A) $[\text{K}^+] = [\text{NO}_3^-] > [\text{Ag}^+] > [\text{SO}_4^{2-}]$
(B) $[\text{K}^+] = [\text{NO}_3^-] > [\text{SO}_4^{2-}] > [\text{Ag}^+]$
(C) $[\text{K}^+] > [\text{NO}_3^-] > [\text{Ag}^+] > [\text{SO}_4^{2-}]$
(D) $[\text{K}^+] > [\text{NO}_3^-] > [\text{SO}_4^{2-}] > [\text{Ag}^+]$
19. An amino acid ($M = 174$) with a formula of $\text{C}_a\text{H}_b\text{N}_x\text{O}_2$ contains 8.1% by mass of H and 41.3% by mass of C. What is the value of *x* in its formula?
- (A) 1 (B) 2 (C) 3 (D) 4
20. Potassium tris(oxalato)ferrate(III) trihydrate is prepared by precipitation from aqueous solution according to the following equation:
- $$3 \text{K}_2\text{C}_2\text{O}_4(aq) + \text{FeCl}_3(aq) + 3 \text{H}_2\text{O}(l) \rightarrow \text{K}_3\text{Fe}(\text{C}_2\text{O}_4)_3 \cdot 3\text{H}_2\text{O}(s) + 3 \text{KCl}(aq)$$
- A student mixes 10.00 g $\text{K}_2\text{C}_2\text{O}_4$ ($M = 166.2$) and 5.00 g FeCl_3 ($M = 162.2$) in 20.0 mL water and isolates 5.74 g $\text{K}_3\text{Fe}(\text{C}_2\text{O}_4)_3 \cdot 3\text{H}_2\text{O}$ by filtration. What is the percent yield of this reaction?
- (A) 19.4% (B) 37.9% (C) 58.3% (D) 65.5%
21. Carbon tetrachloride (CCl_4 , $M = 153.8$, density = 1.59 g mL^{-1}) and heptane (C_7H_{16} , $M = 100.2$, density = 0.68 g mL^{-1}) are miscible liquids. Which solution has the highest concentration of CCl_4 ?
- (A) 5% v/v CCl_4 in heptane
(B) 5% w/w CCl_4 in heptane
(C) 5 mol% CCl_4 in heptane
(D) All three solutions have the same concentration of CCl_4 .

Stoichiometry/Solutions

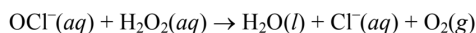
22. A solution of 5.00 g of which substance dissolved in 100.0 g water has the highest boiling point?

(A) H_3PO_4 (B) H_2SO_4
(C) H_3AsO_4 (D) H_2SeO_4

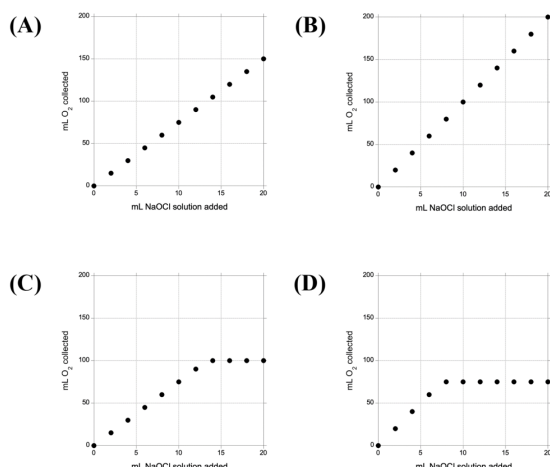
23. A 1.802-g sample of a hydrated double salt containing two different metal ions with the formula of $\text{M}_\text{A}(\text{M}_\text{B})_2(\text{SO}_4)_4 \cdot 22\text{H}_2\text{O}$ is heated strongly to drive off all the water, leaving a residue of 1.000 g. What is the identity of M_A ?

(A) Ca (B) Ti (C) Fe (D) Ba

24. Hypochlorite ion reacts with hydrogen peroxide to produce oxygen gas.



A series of experiments is carried out where 10.0 mL of a 0.30 M solution of hydrogen peroxide is mixed with a varying volume of 0.40 M sodium hypochlorite solution, and the oxygen gas evolved is collected over water. Which graph best represents how the volume of O_2 varies with the volume of NaOCl solution used?



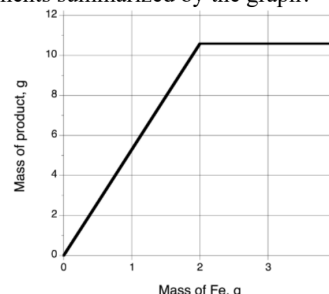
25. 0.125 g of strontium metal reacts with excess liquid ammonia to form strontium amide and hydrogen gas. How many moles of hydrogen are produced?

(A) 7.13×10^{-4} mol (B) 1.43×10^{-3} mol
(C) 2.14×10^{-4} mol (D) 2.85×10^{-3} mol

26. A diamagnetic compound is 71.4% arsenic by mass. What is its molecular formula?

(A) CH_3AsN (B) $\text{C}_2\text{H}_6\text{As}$
(C) $\text{C}_2\text{H}_6\text{As}_2\text{N}_2$ (D) $\text{C}_4\text{H}_{12}\text{As}_2$

27. Weighed samples of iron are added in increments to a given amount of liquid bromine and allowed to react completely. The reaction produces a single product that is isolated and weighed. Which statement best describes the experiments summarized by the graph?

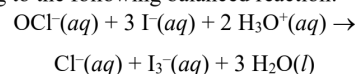


- (A) When 1.00 g of Fe is added, Fe is the limiting reactant.
(B) When 2.00 g of Fe is added, 10.6 g of FeBr_2 is formed.
(C) When 2.50 g of Fe is added, both reactants are used completely.
(D) When 3.50 g of Fe is added, there is an excess of Br_2 .

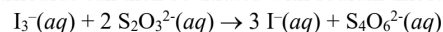
28. A copper ore consists of a mixture of Cu_2S and CuS . An 89.0 g sample of this ore is smelted to produce 67.5 g of elemental copper. What is the mass of CuS in the ore sample?

(A) 13.3 g (B) 26.6 g (C) 37.5 g (D) 64.1 g

29. The hypochlorite concentration in bleach can be determined by treating it with excess iodide ion in acidic solution, which causes the formation of triiodide ion according to the following balanced reaction:



The triiodide can then be titrated with sodium thiosulfate:



A 75.0-mL sample of liquid bleach is treated with excess KI and acid, and then titrated with 0.0235 M sodium thiosulfate solution. The endpoint is observed on the addition of 21.23 mL of this solution. What is the hypochlorite concentration in the bleach?

(A) 4.99×10^{-4} M (B) 1.66×10^{-3} M
(C) 3.33×10^{-3} M (D) 6.65×10^{-3} M

30. Which aqueous solution has the highest normal boiling point?

(A) 0.10 M Na_3PO_4 (B) 0.30 M Na_2HPO_4
(C) 0.50 M NaH_2PO_4 (D) 0.70 M H_3PO_4

Stoichiometry/Solutions

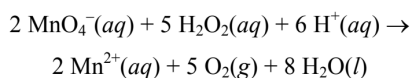
31. Hydrogen and oxygen gas are combined in a reaction vessel and reacted completely to form water. What volumes of gases (at STP) will give the greatest mass of water?
- (A) 1.0 L H₂ and 4.0 L O₂
(B) 2.0 L H₂ and 3.0 L O₂
(C) 3.0 L H₂ and 2.0 L O₂
(D) 4.0 L H₂ and 1.0 L O₂
32. Potassium hydrogen phthalate (KHP, $M = 204$) is a monoprotic acid used as a primary standard in acid-base titrations. It takes 28.15 mL of a solution of Sr(OH)₂ to neutralize 0.402 g KHP. What is the concentration of the Sr(OH)₂ solution?
- (A) 0.00197 M (B) 0.0350 M
(C) 0.0700 M (D) 0.100 M
33. 20.0 mL of 0.15 M aqueous Pb(NO₃)₂ and 10.0 mL of 0.35 M aqueous NaBr are mixed, giving a white precipitate. Which ion is LEAST abundant in the supernatant?
- (A) Pb²⁺ (B) Na⁺ (C) NO₃⁻ (D) Br⁻
34. A sample of 9.736 g of a lead oxide is reduced completely with excess carbon monoxide to give metallic lead. The resulting gas is bubbled into saturated lime water, resulting in the precipitation of 6.636 g calcium carbonate ($M = 100.09$) after drying. What is the formula of this lead oxide?
- (A) PbO (B) Pb₂O₃
(C) Pb₃O₄ (D) Pb₁₂O₁₉
35. A mixture of K₂S, K₂SO₃, and K₂SO₄ is found to contain 60.0% K by mass. What is the mass percent of oxygen in the mixture?
- (A) 15.4% (B) 22.3% (C) 30.3% (D) 36.7%
36. A solution of 5.00 g oxalic acid (H₂C₂O₄, $M = 90.04$) in 100.0 g H₂O has a freezing point of -1.31 °C. What is the percent ionization of oxalic acid in this solution? The freezing point depression constant of water is $K_f = 1.86\text{ }^{\circ}\text{C}\cdot\text{m}^{-1}$.
- (A) 13.5% (B) 26.8% (C) 70.4% (D) 100%
37. Iron(III) sulfide reacts with gaseous hydrogen chloride to produce iron(III) chloride and hydrogen sulfide. How much iron(III) sulfide would be required to react with 1.5 mol HCl?
- (A) 0.25 mol (B) 0.50 mol
(C) 0.75 mol (D) 1.0 mol
38. How much CH₄ remains after 10.0 g of CH₄ reacts completely with 15.0 g of O₂ according to the following equation?
- $$\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{g})$$
- (A) 0.00 g (B) 2.50 g (C) 5.00 g (D) 6.25 g
39. A sample of 1.00 g of which compound contains the greatest number of atoms?
- (A) CO ($M = 28.0$) (B) C₂H₆ ($M = 30.0$)
(C) C₆H₆ ($M = 78.0$) (D) Kr ($M = 83.8$)
40. 200.0 mL of aqueous nitric acid is mixed with 300.0 mL of 0.200 M Mg(NO₃)₂ solution to give a solution with a nitrate ion concentration of 0.500 M. What is the concentration of the nitric acid?
- (A) 0.100 M (B) 0.650 M
(C) 0.950 M (D) 1.25 M
41. A student wishes to prepare a solution with a final concentration of Na⁺ = 0.50 M and a final concentration of HCO₃⁻ = 0.10 M by taking some NaOH and some trona (Na₂CO₃•NaHCO₃•2H₂O) and diluting with water to a final volume of 1.00 L. How much NaOH and trona are required?
- (A) 0.05 mol NaOH, 0.15 mol trona
(B) 0.20 mol NaOH, 0.10 mol trona
(C) 0.35 mol NaOH, 0.05 mol trona
(D) A solution cannot be prepared with the desired concentrations of Na⁺ and HCO₃⁻ using only NaOH and trona.
42. As the concentration of an aqueous solution of sodium chloride increases from 1.0 M to 2.0 M, the values of which properties of the solution decrease?
- I. Density
II. Freezing point
- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II
43. Which sample contains the greatest number of molecules?
- (A) 35.0 g C₂H₂ (B) 45.0 g C₂H₆
(C) 60.0 g C₄H₁₀ (D) 100.0 g C₆H₆
44. A mixture of hydrogen and oxygen is ignited and reacts completely to form water. Which mixture will NOT produce 18 g of H₂O?
- (A) 2.0 g H₂, 16.0 g O₂ (B) 2.0 g H₂, 18.0 g O₂
(C) 4.0 g H₂, 16.0 g O₂ (D) 4.0 g H₂, 18.0 g O₂

Stoichiometry/Solutions

45. How many oxygen atoms are in 1.00 g of the mineral troegerite, $(\text{UO}_2)_3(\text{AsO}_4)_2 \cdot 12 \text{H}_2\text{O}$ ($M = 1304.0$)?
- (A) 6.47×10^{21} (B) 8.31×10^{21}
(C) 1.20×10^{22} (D) 1.39×10^{22}
46. Hünig's base is a monobasic amine containing 74.34% C, 14.82% H, and 10.84% N by mass. How many hydrogen atoms are in one molecule of Hünig's base?
- (A) 8 (B) 14 (C) 15 (D) 19
47. Magnesium metal burns in air to form a mixture of magnesium oxide (MgO , $M = 40.31$) and magnesium nitride (Mg_3N_2 , $M = 100.95$). A 1.000 g sample of magnesium ribbon is burned in air to give 1.584 g of the oxide/nitride mixture. What percentage of the magnesium is present in the form of the nitride?
- (A) 9.00% (B) 11.0%
(C) 27.1% (D) 90.3%
48. What is the pH of a solution formed by mixing 45.0 mL of 0.10 M HNO_3 , 50.0 mL of 0.20 M HCl , and 55.0 mL of 0.10 M CH_3COOH ?
- (A) 0.40 (B) 0.88 (C) 1.01 (D) 1.52
49. Upon heating in a stream of hydrogen gas, 0.688 g of a manganese oxide is reduced to metallic manganese and forms 0.235 g water. What is the formula of the oxide?
- (A) MnO (B) Mn_2O_3 (C) Mn_3O_4 (D) MnO_2
50. Which 1.00 g sample of an element contains the greatest number of molecules?
- (A) Buckminsterfullerene, C_{60}
(B) Ozone, O_3
(C) White phosphorus, P_4
(D) Sulfur, S_8
51. Hydrogen gas reacts with 5.00 L of nitrogen gas under constant temperature and pressure conditions to form 10.0 L of ammonia gas. What is the minimum volume of hydrogen gas required to produce this amount of ammonia?
- (A) 5.00 L (B) 7.50 L (C) 15.0 L (D) 30.0 L
52. Phosphoric acid can be manufactured according to the following reaction:
- $$\text{Ca}_3(\text{PO}_4)_2 + 3 \text{SiO}_2 + 5 \text{C} + 5 \text{O}_2 + 3 \text{H}_2\text{O} \rightarrow 3 \text{CaSiO}_3 + 5 \text{CO}_2 + 2 \text{H}_3\text{PO}_4$$
- If *equal masses* of calcium phosphate ($M = 310.$) and silica ($M = 60.0$) are reacted with excess carbon, oxygen, and water to produce 1.00×10^3 kg phosphoric acid ($M = 98.0$), what mass of calcium phosphate was used, assuming 100% yield?
- (A) 610. kg (B) 800. kg
(C) 920. kg (D) 1580 kg
53. After mixing 30.0 mL of 0.30 M $\text{Ca}(\text{NO}_3)_2$ solution and 15.0 mL of 0.60 M NaF solution, which ions are present in solution at concentrations of at least 0.10 M?
- I. Ca^{2+} II. F^-
- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II
54. A salt whose formula is $\text{Na}_2\text{S}_x\text{O}_y$ is 47.5% sulfur by mass. What is the value of x in the formula?
- (A) 1 (B) 2 (C) 3 (D) 4
55. Lithium reacts with water to form lithium hydroxide. What mass of lithium is required to produce 12 g of lithium hydroxide?
- (A) 2.0 g (B) 3.5 g (C) 7.0 g (D) 12 g
56. Complete combustion of 1.00 g of the hydrocarbon pagodane gives 3.38 g carbon dioxide. What is the empirical formula of pagodane?
- (A) CH (B) CH_2 (C) C_2H_5 (D) C_3H_8
57. Electrolysis of 10.00 g of a binary metal chloride deposits 6.207 g of the pure metal. What is the metal?
- (A) Cu ($Z = 29$) (B) Cd ($Z = 48$)
(C) Ce ($Z = 58$) (D) Th ($Z = 90$)
58. Decomposition of 1.0 g of which of the following compounds into its constituent elements gives the greatest amount of N_2 gas?
- (A) NO (B) NO_2 (C) N_2O_4 (D) NH_3

Stoichiometry/Solutions

59. Permanganate ion oxidizes hydrogen peroxide in acidic solution according to the following equation:



If 35.0 mL of an acidic 0.150 M KMnO_4 solution is required to consume all the H_2O_2 in 50.0 mL of a disinfectant solution, what is the concentration of H_2O_2 in the disinfectant?

- (A) 0.0420 M (B) 0.105 M
(C) 0.263 M (D) 0.368 M

60. A solution containing 10 g of which substance dissolved in 100 g of water will show the greatest freezing point depression compared to pure water?

- (A) Anhydrous magnesium sulfate, MgSO_4
(B) Magnesium sulfate heptahydrate, $\text{MgSO}_4 \cdot 7 \text{H}_2\text{O}$
(C) Anhydrous sodium thiosulfate, $\text{Na}_2\text{S}_2\text{O}_3$
(D) Sodium thiosulfate pentahydrate, $\text{Na}_2\text{S}_2\text{O}_3 \cdot 5 \text{H}_2\text{O}$

61. One mole of which hydrocarbon requires 8 mol O_2 to achieve complete combustion to give carbon dioxide and water?

- (A) C_3H_8 (B) C_4H_{10} (C) C_5H_{10} (D) C_5H_{12}

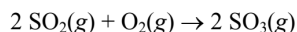
62. What is the concentration of calcium ions in 350. mL of an aqueous solution containing 7.50 g CaCl_2 ?

- (A) 0.0676 M (B) 0.193 M
(C) 0.284 M (D) 0.535 M

63. 0.422 g of an element Z reacts with oxygen to form 0.797 g of the oxide Z_2O_3 . What is the element Z?

- (A) Al (B) Sc (C) Cr (D) Ga

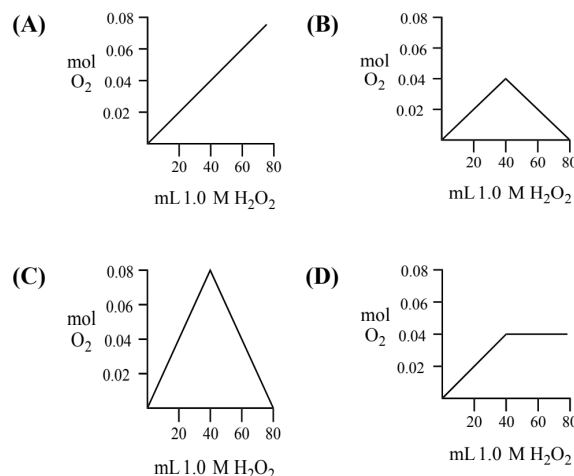
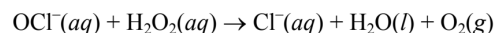
64. At constant temperature and pressure, 5.0 L of SO_2 is combined with 3.0 L of O_2 according to the equation:



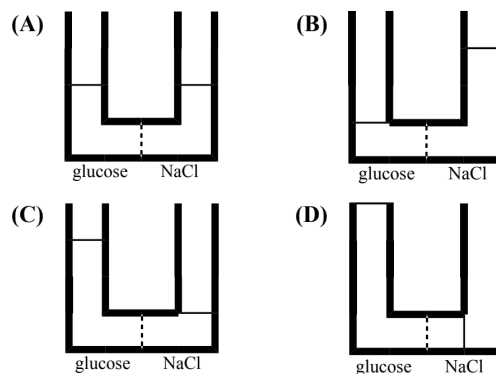
After SO_3 formation is complete, what is the volume of the mixture?

- (A) 5.5 L (B) 6.0 L (C) 6.5 L (D) 8.0 L

65. A sample of 0.040 mol hypochlorite ion is treated with varying amounts of 1.0 M aqueous H_2O_2 . Which graph represents the amount of $\text{O}_2(g)$ that is evolved according to the chemical reaction shown?



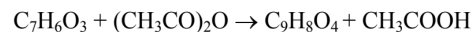
66. Equal volumes of aqueous 1.00 *m* glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) and 1.00 *m* sodium chloride solutions are placed on opposite sides of a U-tube, separated by a semipermeable membrane (through which only water can diffuse). What will the setup look like at equilibrium?



67. Calcium carbonate, CaCO_3 , decomposes upon heating to calcium oxide and carbon dioxide. What mass of solid calcium carbonate is required to produce 2.40 liters of carbon dioxide measured at STP?

- (A) 10.7 g (B) 21.4 g (C) 50.0 g (D) 100. g

68. Aspirin, $\text{C}_9\text{H}_8\text{O}_4$, is prepared by the acetylation of salicylic acid, $\text{C}_7\text{H}_6\text{O}_3$, according to the following equation:



If the yield of this reaction is 83%, what mass of salicylic acid would be required to prepare 1.0 kg of aspirin?

- (A) 0.77 kg (B) 0.92 kg
(C) 1.2 kg (D) 1.3 kg

Stoichiometry/Solutions

69. A solution is prepared by mixing 25.0 mL of 6.0 M HCl with 45.0 mL of 3.0 M HNO₃. What is [H⁺] in the resulting solution?
(A) 1.9 M (B) 2.1 M (C) 4.1 M (D) 4.5 M
70. A 5.73 g sample of a liquid hydrocarbon burned in excess oxygen produces 17.48 g CO₂. What is the formula of the hydrocarbon?
(A) C₅H₁₂ (B) C₆H₆
(C) C₆H₁₀ (D) C₆H₁₂
71. A solution of 3.00 g of which substance, dissolved in 100 g H₂O, has the highest boiling point?
(A) HOCH₂CH(OH)CH₂OH (*M* = 92.1)
(B) RbF (*M* = 104.5)
(C) AlCl₃ (*M* = 133.3)
(D) TiNO₃ (*M* = 390.4)
72. Boron carbide, B₄C, is made by the high temperature reaction of boron oxide with graphite, yielding carbon monoxide as a by-product.
___ B₂O₃ + ___ C → ___ B₄C + ___ CO
What is the total of the smallest coefficients for the reactants and products in the balanced equation?
(A) 9 (B) 10 (C) 15 (D) 16
73. A compound with 69.41% C, 4.16% H and 26.42% O has a molar mass of 230 – 250 g•mol⁻¹. What is its molecular formula?
(A) C₁₃H₉O₄ (B) C₁₄H₁₀O₄
(C) C₁₃H₆O₄ (D) C₁₅H₁₄O₃
74. Aluminum reacts with sulfur to form aluminum sulfide. If 31.9 g of Al are reacted with 72.2 g of S, what is the theoretical yield of aluminum sulfide in grams?
(A) 88.8 g (B) 69.7 g (C) 57.2 g (D) 113 g
75. According to EPA guidelines the permissible level for lead in drinking water is 15 parts per billion (ppb). What is the maximum allowable mass of lead that could be present in 1.00 L of H₂O?
(A) 0.015 ng (B) 0.015 µg
(C) 0.015 mg (D) 0.015 g
76. Aqueous solutions of electrolytes that produce two ions upon ionization typically exhibit freezing point depressions that are twice as large as solutions of non-electrolytes with the same concentration. Which 0.10 M electrolyte solution will show the smallest depression?
(A) HCl (B) NaBr (C) KNO₃ (D) MgSO₄
77. Interferon is a water-soluble protein. A solution prepared by dissolving 15.0 mg of interferon in 2.50 mL of H₂O exhibits an osmotic pressure of 5.80 mm Hg at 25 °C. What is the molar mass of interferon?
(A) 1.92 × 10⁴ g•mol⁻¹ (B) 1.92 × 10⁷ g•mol⁻¹
(C) 1.95 × 10⁶ g•mol⁻¹ (D) 1.61 × 10³ g•mol⁻¹
78. A reaction mixture is prepared containing 0.60 mol of aluminum and 1.20 mol of manganese dioxide. The mixture is heated until one of the reactants has been completely consumed according to the equation:
2 Al + 3 MnO₂ → 3 Mn + Al₂O₃
What quantity of which reactant remains uncombined?
(A) 0.20 mol Al (B) 0.40 mol Al
(C) 0.30 mol MnO₂ (D) 0.60 mol MnO₂
79. A solution with a mass of 1.263 g containing an unknown amount of potassium ions was treated with excess sodium tetraphenylborate to precipitate 1.003 g of KB(C₆H₅)₄ (*M* = 358.33). What is the mass percentage of potassium in the original solution?
(A) 8.67% (B) 9.16% (C) 10.9% (D) 13.8%
80. A typical polyethylene grocery bag weighs 12.4 g. How many metric tons of CO₂ would be released into the atmosphere if the 102 billion bags used in one year in the United States were burned? [1 metric ton = 1000 kg]
(A) 4.52 × 10⁴ (B) 1.99 × 10⁶
(C) 3.98 × 10⁶ (D) 3.98 × 10⁹
81. Which mixture of water and H₂SO₄ represents a solution with a concentration that is closest to 30% by mass H₂SO₄?
(A) 30 g H₂SO₄ + 100 g H₂O
(B) 1 mol H₂SO₄ + 200 g H₂O
(C) 30 mol H₂SO₄ + 0.70 kg H₂O
(D) 0.30 mol H₂SO₄ + 0.70 mol H₂O

Stoichiometry/Solutions

82. How would the freezing point depression of a 0.05 *m* CaCl₂ solution compare with that of a NaCl solution? It would be
- (A) less than that for a 0.10 *m* NaCl solution.
(B) between that for a 0.10 *m* NaCl solution and a 0.20 *m* NaCl solution.
(C) between that for a 0.20 *m* NaCl solution and a 0.30 *m* NaCl solution.
(D) greater than that for a 0.30 *m* NaCl solution.
83. What is the mole fraction of CH₃OH in an aqueous solution that is 12.0 *m* in CH₃OH?
- (A) 0.178 (B) 0.216 (C) 0.400 (D) 0.667
84. Three different oxides of nitrogen each contain 7.00 g of nitrogen and weigh 15.0 g, 23.0 g and 19.0 g, respectively. What are their empirical formulas?
- (A) NO, NO₂, N₂O₃ (B) NO, N₂O₃, N₂O₅
(C) N₂O, NO₂, N₂O₅ (D) NO₂, N₂O₃, N₂O₅
85. Fe₂O₃ reacts with excess CO at a high temperature according to the equation below.
- $$\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Fe} + 3\text{CO}_2$$
- If 6.50 g of Fe₂O₃ yields 3.85 g of Fe what is the percentage yield of the reaction?
- (A) 59.2% (B) 69.9% (C) 76.3% (D) 84.7%
86. What is the final [Na⁺] in a solution prepared by mixing 70.0 mL of 3.00 M Na₂SO₄ with 30.0 mL of 1.00 M NaCl?
- (A) 2.00 M (B) 2.40 M (C) 4.00 M (D) 4.50 M
87. The mass percentage of O in a potassium salt, K₂S₂O_x, is 36.0%. What is the formula of the polyatomic ion?
- (A) S₂O₃²⁻ (B) S₂O₅²⁻ (C) S₂O₇²⁻ (D) S₂O₈²⁻
88. Cu reacts with HNO₃ according to the equation
- $$\text{Cu} + \text{HNO}_3 \rightarrow \text{Cu}(\text{NO}_3)_2 + \text{NO} + \text{NO}_2 + \text{H}_2\text{O}$$
- If NO and NO₂ are formed in a 2:3 ratio, what is the coefficient for Cu when the equation is balanced with the simplest whole numbers?
- (A) 2 (B) 3 (C) 6 (D) 9
89. The active ingredient in commercial bleach is sodium hypochlorite, NaOCl, which can be determined by iodometric analysis as indicated in these equations.
- $$\text{OCl}^- + 2\text{H}^+ + 2\text{I}^- \rightarrow \text{I}_2 + \text{Cl}^- + \text{H}_2\text{O}$$
- $$\text{I}_2 + 2\text{S}_2\text{O}_3^{2-} \rightarrow \text{S}_4\text{O}_6^{2-} + 2\text{I}^-$$
- If 1.356 g of a bleach sample requires 19.50 mL of 0.100 M Na₂S₂O₃ solution, what is the percentage by mass of NaOCl in the bleach?
- (A) 2.68% (B) 3.70% (C) 5.35% (D) 10.7%
90. A 12.0 M acid solution that contains 75.0% acid by mass has a density of 1.57 g/mL. What is the identity of the acid?
- (A) HCl (*M* = 36.5) (B) CH₃CO₂H (*M* = 60.0)
(C) HBr (*M* = 80.9) (D) H₃PO₄ (*M* = 98.0)
91. A 10.00 g sample of a compound containing C, H, and O is burned completely to produce 14.67 g of CO₂ and 6.000 g of H₂O. What is the empirical formula of this compound?
- (A) CHO (B) CH₂O (C) CH₂O₂ (D) C₂H₄O
92. What is the molality of a solution made by dissolving 36.0 g of glucose (C₆H₁₂O₆, *M* = 180.2) in 64.0 g of H₂O?
- (A) 0.0533 (B) 0.200 (C) 0.360 (D) 3.12
93. Which aqueous solution freezes at the lowest temperature?
- (A) 0.30 *m* C₂H₅OH (B) 0.25 *m* KNO₃
(C) 0.20 *m* CaBr₂ (D) 0.10 *m* FeCl₃
94. Element E reacts with oxygen to produce EO₂. Identify element E if 16.5 g of it react with excess oxygen to form 26.1 g of EO₂.
- (A) manganese (B) nickel
(C) sulfur (D) titanium
95. Ethanol (C₂H₅OH, *M* = 46) and methanol (CH₃OH, *M* = 32) form an ideal solution when mixed. What is the vapor pressure of a solution prepared by mixing equal masses of ethanol and methanol? (The vapor pressures of ethanol and methanol are 44.5 mm Hg and 88.7 mm Hg, respectively.)
- (A) 133 mm Hg (B) 70.6 mm Hg
(C) 66.6 mm Hg (D) 44.5 mm Hg
96. What is the molarity of Na⁺ ions in a solution made by dissolving 4.20 g of NaHCO₃ (*M* = 84.0) and 12.6 g of Na₂CO₃ (*M* = 126) in water and diluting to 1.00 L?
- (A) 0.050 M (B) 0.100 M
(C) 0.150 M (D) 0.250 M

Stoichiometry/Solutions

97. The percent composition of the high explosive HNS is

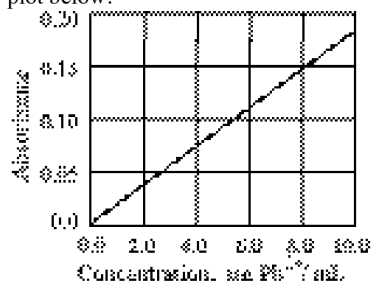
C	H	N	O
37.35%	1.34%	18.67%	42.65%

The molar mass of HNS is 450.22. What is the molecular formula of HNS?

- (A) $C_{13}H_4N_7O_{12}$ (B) $C_{14}H_6N_6O_{12}$
 (C) $C_{15}H_{10}N_6O_{11}$ (D) $C_{16}H_{12}N_5O_{11}$
98. A student prepares four 0.10 M solutions, each containing one of the solutes below. Which solution has the lowest freezing point?
- (A) $CaCl_2$ (B) KOH
 (C) $NaC_2H_3O_2$ (D) NH_4NO_3
99. What is the molarity of a hydrochloric acid solution if 20.00 mL of it neutralizes 18.46 mL of a 0.0420 M $Ba(OH)_2$ solution?
- (A) 0.0194 M (B) 0.0388 M
 (C) 0.0455 M (D) 0.0775 M

100. What is the mass percent of oxygen in the compound $UO_2(C_2H_3O_2)_2 \cdot NH_4C_2H_3O_2 \cdot 6H_2O$?
- | Formula Weight, g |
|-------------------|
| 573 |
- (A) 5.58% (B) 16.8% (C) 22.3% (D) 39.1%

101. A 25.0 mL sample of waste water is obtained to analyze for Pb^{2+} ions. This sample is evaporated to dryness and redissolved in 2.0 mL of H_2O , mixed with 2.0 mL of a buffer solution and 2.0 mL of a solution of dithizone then diluted to 10.0 mL. The absorbance of the colored Pb^{2+} -dithizone complex is compared with the Beer-Lambert plot below.



The absorbance of a portion of the final solution is 0.13. What is the concentration of Pb^{2+} ions in the waste water in ppm?

- (A) 2.9 (B) 7.2 (C) 18 (D) 36
102. Reductic acid contains 52.63% carbon, 5.30% hydrogen, and 42.07% oxygen. Its empirical formula is the same as its molecular formula. What is the number of carbon atoms in a molecule of this acid?
- (A) 4 (B) 5 (C) 6 (D) 8

103. Sulfur trioxide, SO_3 , is made by oxidizing sulfur dioxide, SO_2 , according to the equation, $2SO_2 + O_2 \rightarrow 2SO_3$. If a 16.0 g sample of SO_2 yields 18.0 g of SO_3 , what is the percent yield?

(A) 70.0% (B) 80.0% (C) 90.0% (D) 100%

104. A 1.0 g sample of which substance contains the largest number of molecules?

(A) HN_3 (B) N_2H_4 (C) H_2O_2 (D) HCl

105. What is the maximum mass of PbI_2 that can be precipitated by mixing 25.0 mL of 0.100 M $Pb(NO_3)_2$ with 35.0 mL of 0.100 M NaI ?

Molar Mass, $g \cdot mol^{-1}$	
PbI_2	461

(A) 0.807 g (B) 1.15 g (C) 1.61 g (D) 2.30 g

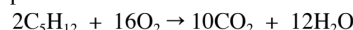
106. Which aqueous solution has the highest osmotic pressure at 25°C? (Assume all ionic compounds ionize completely in solution.)

(A) 0.1 M $Al_2(SO_4)_3$ (B) 0.1 M Na_2CO_3
 (C) 0.2 M $KMnO_4$ (D) 0.3 M $C_6H_{12}O_6$

107. The mass of a single molecule of an allotrope of sulfur is 3.20×10^{-22} g. How many sulfur atoms are present in a molecule of this allotrope?

(A) 4 (B) 6 (C) 8 (D) 12

108. 100. L of carbon dioxide measured at 740. mmHg and 50 °C is produced by the complete combustion of a sample of pentane.



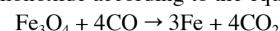
What mass of pentane reacted?

(A) 342 g (B) 265 g (C) 64.4 g (D) 53.0 g

109. Which 0.10 M aqueous solution has the smallest change in freezing point relative to pure water?

(A) $HC_2H_3O_2$ (B) HCl
 (C) $CaCl_2$ (D) $AlCl_3$

110. Magnetite, Fe_3O_4 , can be reduced to iron by heating with carbon monoxide according to the equation:



What mass of Fe_3O_4 is required in order to obtain 5.0 kg of iron if the process is 88% efficient?

(A) 6.1 kg (B) 6.9 kg (C) 7.9 kg (D) 18 kg

Stoichiometry/Solutions

111. 40.0 g of a solute is dissolved in 500. mL of a solvent to give a solution with a volume of 515 mL. The solvent has a density of 1.00 g/mL. Which statement about this solution is correct?
- (A) The molarity is greater than the molality.
 (B) The molarity is lower than the molality.
 (C) The molarity is the same as the molality.
 (D) The molarity and molality cannot be compared without knowing the solute.
112. What is the molarity of a 0.500 molal aqueous solution of calcium nitrate that has a density of 1.045 g·mL⁻¹?
- (A) 0.483 M (B) 0.500 M
 (C) 0.522 M (D) 0.567 M
113. What volume of 0.150 M H₂SO₄ would be required to completely neutralize a mixture of 20.0 mL of 0.200 M NaOH and 40.0 mL of 0.0500 M Ca(OH)₂?
- (A) 20.0 mL (B) 26.7 mL
 (C) 40.0 mL (D) 53.3 mL
114. A compound with the formula X₂O₅ contains 34.8% oxygen by mass. Identify element X.
- (A) arsenic (B) carbon
 (C) phosphorous (D) samarium
115. A solution of 0.0400 mol of C₂H₄Br₂ and 0.0600 mol of C₃H₆Br₂ exerts a vapor pressure of 145.4 mm Hg at a certain temperature. Determine the vapor pressure of pure C₃H₆Br₂ at this temperature. Assume the vapor pressure of C₂H₄Br₂ at this temperature is 173 mm Hg and that the solution obeys Raoult's Law.
- (A) 76.2 mm Hg (B) 118 mm Hg
 (C) 127 mm Hg (D) 138 mm Hg
116. When 0.1 M aqueous solutions of aluminum nitrate, magnesium nitrate, sodium nitrate and urea, (NH₂)₂CO, are arranged in order of increasing boiling point, which order is correct?
- (A) Al(NO₃)₃ = Mg(NO₃)₂ = (NH₂)₂CO = NaNO₃
 (B) Mg(NO₃)₂ < (NH₂)₂CO < NaNO₃ < Al(NO₃)₃
 (C) (NH₂)₂CO < NaNO₃ < Mg(NO₃)₂ < Al(NO₃)₃
 (D) NaNO₃ < Mg(NO₃)₂ < Al(NO₃)₃ < (NH₂)₂CO
117. What is the maximum mass of Ba₃(PO₄)₂ that can be formed from 0.00240 mol of Ba(NO₃)₂ and 0.131 g of Na₃PO₄?
- | Molar Mass / g·mol ⁻¹ | |
|---|--------|
| Ba ₃ (PO ₄) ₂ | 601.84 |
| Na ₃ PO ₄ | 163.94 |
- (A) 0.240 g (B) 0.480 g
 (C) 1.44 g (D) 7.22 g
118. What is the molarity of KI in a solution that is 5.00% KI by mass and has a density of 1.038 g·cm⁻³?
- (A) 0.0301 M (B) 0.313 M
 (C) 0.500 M (D) 0.625 M
119. What is the concentration of the solution that results from mixing 40.0 mL of 0.200 M HCl with 60.0 mL of 0.100 M NaOH? (You may assume the volumes are additive.)
- (A) 0.150 M NaCl
 (B) 0.0200 M NaCl and 0.0200 M HCl
 (C) 0.0200 M NaCl and 0.0600 M HCl
 (D) 0.0600 M NaCl and 0.0200 M HCl
120. Mole fractions are typically used to calculate which properties for solutions containing nonvolatile solutes?
- I. freezing point depression
 II. osmotic pressure
 III. vapor pressure
- (A) I only (B) III only
 (C) I and II only (D) II and III only
121. A 1.0 L portion of a 0.30m solution of which of the following would be most effective at removing ice from a sidewalk?
- (A) C₆H₁₂O₆ (B) NaBr
 (C) KNO₃ (D) CaCl₂
122. C₆H₆ + Br₂ ⇌ C₆H₅Br + HBr
 In an experiment to prepare bromobenzene according to the equation, a student reacted 20.0 g of C₆H₆ with 0.310 mol of bromine. If 28.0 g of C₆H₅Br was obtained, what was the percentage yield?
- (A) 31.5 (B) 40.3 (C) 57.6 (D) 69.7
123. Methylamine, CH₃NH₂, reacts with O₂ to form CO₂, N₂, and H₂O. What amount of O₂ (in moles) is required to react completely with 1.00 mol of CH₃NH₂?
- (A) 2.25 (B) 2.50 (C) 3.00 (D) 4.50
124. Iodine adds to the double bonds in fatty acids (one iodine molecule per double bond). How many double bonds are in a molecule of arachidonic acid (Molar mass = 304.5 g/mol) if 0.125 g of the acid require 0.417 g of iodine?
- (A) 2 (B) 3 (C) 4 (D) 8
125. A mineral containing only manganese and oxygen contains 69.6% Mn by mass. What is its empirical formula?
- (A) MnO (B) Mn₂O₃
 (C) Mn₃O₄ (D) MnO₂

Stoichiometry/Solutions

126. Toluene, C_7H_8 , is added to gasoline to increase its octane rating. What is the volume ratio of air to toluene vapor to burn completely to form CO_2 and H_2O ? (Assume air is 20% O_2 by volume.)
 (A) 9/1 (B) 11/1 (C) 28/1 (D) 45/1
127. Acidified solutions of dichromate ion, $Cr_2O_7^{2-}$, oxidize Fe^{2+} to Fe^{3+} , forming Cr^{3+} in the process. What volume of 0.175 M $K_2Cr_2O_7$ in mL is required to oxidize 60.0 mL of 0.250 M $FeSO_4$?
 (A) 14.3 (B) 28.6 (C) 42.9 (D) 85.7
128. A 1.871 gram sample of an unknown metallic carbonate is decomposed by heating to form the metallic oxide and 0.656 g of carbon dioxide according to the equation

$$MCO_3(s) \rightarrow MO(s) + CO_2(g)$$

 What is the metal?
 (A) Ca (B) Mn (C) Ni (D) Zn
129. An ionic compound contains 29.08% sodium, 40.56% sulfur and 30.36% oxygen by mass. What is the formula of the sulfur-containing anion in the compound?
 (A) $S_2O_3^{2-}$ (B) $S_2O_4^{2-}$
 (C) $S_2O_5^{2-}$ (D) $S_2O_6^{2-}$
130. A solution is prepared containing a 2:1 mol ratio of dibromoethane ($C_2H_4Br_2$) and dibromopropane ($C_3H_6Br_2$). What is the total vapor pressure over the solution assuming ideal behavior?
- | Vapor pressure (mmHg) | |
|-----------------------|-----|
| $C_2H_4Br_2$ | 173 |
| $C_3H_6Br_2$ | 127 |
- (A) 300 mmHg (B) 158 mmHg
 (C) 150 mmHg (D) 142 mmHg
131. A solution of magnesium chloride that is 5.10% magnesium by mass has a density 1.17 g/mL. How many moles of Cl^- ions are in 300. mL of the solution?
 (A) 0.368 (B) 0.627 (C) 0.737 (D) 1.47
132. Which aqueous solution has a freezing point closest to that of 0.30 M $C_{12}H_{22}O_{11}$?
 (A) 0.075 M $AlCl_3$ (B) 0.15M $CuCl_2$
 (C) 0.30 M $NaCl$ (D) 0.60 M $C_6H_{12}O_6$
133. The mass percentages in a compound are carbon 57.48%, hydrogen 4.22% and oxygen 38.29%. What is its empirical formula?
 (A) C_2H_2O (B) $C_4H_3O_2$
 (C) $C_5H_4O_2$ (D) $C_8H_7O_4$
134. Which would produce the largest change in the H_2O level when added to water in a 25 mL graduated cylinder?
 (A) 10.0 g of Hg ($d = 13.6 \text{ g}\cdot\text{mL}^{-1}$)
 (B) 7.42 g of Al ($d = 2.70 \text{ g}\cdot\text{mL}^{-1}$)
 (C) 5.09 g of iron pyrite ($d = 4.9 \text{ g}\cdot\text{mL}^{-1}$)
 (D) 2.68 g of oak ($d = 0.72 \text{ g}\cdot\text{mL}^{-1}$)
135. Diborane, B_2H_6 , can be prepared by the reaction;

$$3NaBH_4 + 4BF_3 \rightarrow 3NaBF_4 + 2B_2H_6$$

 If this reaction has a 70 percent yield, how many moles of $NaBH_4$ should be used with excess BF_3 in order to obtain 0.200 mol of B_2H_6 ?
 (A) 0.200 mol (B) 0.210 mol
 (C) 0.300 mol (D) 0.429 mol
136. What volume of 6.0 M H_2SO_4 should be mixed with 10. L of 1.0 M H_2SO_4 to make 20. L of 3.0 M H_2SO_4 upon dilution to volume?
 (A) 1.7 L (B) 5.0 L (C) 8.3 L (D) 10. L
137. An aqueous solution that is 30.0% NaOH by mass has a density of $1.33 \text{ g}\cdot\text{mL}^{-1}$. What is the molarity of NaOH in this solution?
 (A) 8.25 (B) 9.98 (C) 16.0 (D) 33.2
138. Benzene melts at 5.50°C and has a freezing point depression constant of $5.10^\circ\text{C}\cdot\text{m}^{-1}$. Calculate the freezing point of a solution that contains 0.0500 mole of acetic acid, CH_3COOH , in 125 g of benzene if acetic acid forms a dimer in this solvent.
 (A) 3.46°C (B) 4.48°C
 (C) 5.24°C (D) 6.01°C
139. A solution of sulfuric acid in water that is 25% H_2SO_4 by mass has a density of $1.178 \text{ g}\cdot\text{mL}^{-1}$. Which expression gives the molarity of this solution?
 (A) $0.25 \times 98 \times 1178$ (B) $\frac{0.25 \times 1178}{98}$
 (C) $\frac{0.25}{98 \times 1178}$ (D) $\frac{1178}{0.25 \times 98}$
140. The mineral *trona* has the formula $Na_2CO_3 \cdot NaHCO_3 \cdot 2H_2O$ and a formula mass of $226 \text{ g}\cdot\text{mol}^{-1}$. How many mL of 0.125 M HCl are needed to convert all the carbonate and bicarbonate in a 0.407 g sample of *trona* into carbon dioxide and water?
 (A) 43.2 mL (B) 28.8 mL
 (C) 21.6 mL (D) 14.4 mL

Stoichiometry/Solutions

141. The percentages by mass of C, H, and Cl in a compound are C 52.2%, H 3.7%, and Cl 44.1%. How many carbon atoms are in the simplest formula of the compound?

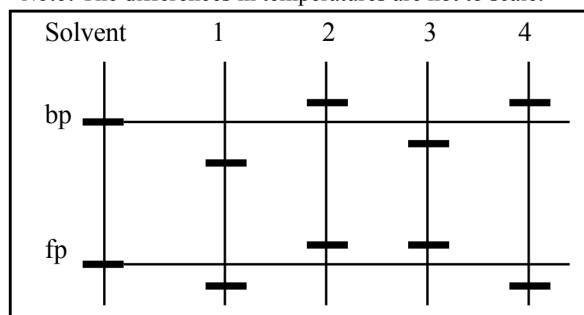
(A) 3 (B) 4 (C) 6 (D) 7

142. $4\text{K}_2\text{O}_2(s) + 2\text{CO}_2(g) \rightarrow 2\text{K}_2\text{CO}_3(s) + 3\text{O}_2(g)$
What is the maximum volume of oxygen that can be produced when 150. mL of CO_2 is passed over 0.500 g of K_2O_2 ? Assume all gases are measured at 0 °C and 1 atm.

(A) 118 mL (B) 157 mL
(C) 225 mL (D) 475 mL

143. The first vertical line in the diagram represents a thermometer with the boiling and freezing points for a pure solvent. The numbered lines represent possible boiling and freezing points for a solution of a nonvolatile solute in the same solvent. Which line best represents the boiling point and freezing point of a solution relative to values for the pure solvent?

Note: The differences in temperatures are not to scale.



(A) 1 (B) 2 (C) 3 (D) 4

144. What mass of magnesium hydroxide is required to neutralize 125 mL of 0.136 M hydrochloric acid solution?

(A) 0.248 g (B) 0.496 g
(C) 0.992 g (D) 1.98 g

Substance	Molar Mass
$\text{Mg}(\text{OH})_2$	$58.33 \text{ g} \cdot \text{mol}^{-1}$

145. Calculate the mass of ammonia that can be produced from the decomposition of a sample of $(\text{NH}_4)_2\text{PtCl}_6$ containing 0.100 g Pt.

(A) 0.0811 g (B) 0.0766 g
(C) 0.0175 g (D) 0.00766 g

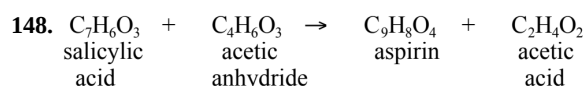
Substance	Molar Mass
$(\text{NH}_4)_2\text{PtCl}_6$	$443.9 \text{ g} \cdot \text{mol}^{-1}$

146. Assume 0.10 L of N_2 and 0.18 L of H_2 , both at 50 atm and 450 °C, are reacted to form NH_3 . Assuming the reaction goes to completion, identify the reagent that is in excess and determine the volume that remains at the same temperature and pressure.

(A) H_2 , 0.02 L (B) H_2 , 0.08 L
(C) N_2 , 0.01 L (D) N_2 , 0.04 L

147. Concentrated hydrochloric acid is 12.0 M and is 36.0% hydrogen chloride by mass. What is its density?

(A) $1.22 \text{ g} \cdot \text{mL}^{-1}$ (B) $1.10 \text{ g} \cdot \text{mL}^{-1}$
(C) $1.01 \text{ g} \cdot \text{mL}^{-1}$ (D) $0.820 \text{ g} \cdot \text{mL}^{-1}$



What is the percent yield if 0.85 g of aspirin is formed in the reaction of 1.00 g of salicylic acid with excess acetic anhydride?

Substance	Molar Mass
$\text{C}_7\text{H}_6\text{O}_3$	$138.12 \text{ g} \cdot \text{mol}^{-1}$
$\text{C}_4\text{H}_6\text{O}_3$	$102.09 \text{ g} \cdot \text{mol}^{-1}$
$\text{C}_9\text{H}_8\text{O}_4$	$180.15 \text{ g} \cdot \text{mol}^{-1}$
$\text{C}_2\text{H}_4\text{O}_2$	$60.05 \text{ g} \cdot \text{mol}^{-1}$

(A) 65 % (B) 77 % (C) 85 % (D) 91 %

149. A 1.50 g sample of an ore containing silver was dissolved, and all of the Ag^+ was converted to 0.124 g of Ag_2S . What was the percentage of silver in the ore?

(A) 6.41% (B) 7.20%
(C) 8.27% (D) 10.8%

150. Methyl-*t*-butyl ether, $\text{C}_5\text{H}_{12}\text{O}$, is added to gasoline to promote cleaner burning. How many moles of oxygen gas, O_2 , are required to burn 1.0 mol of this compound completely to form carbon dioxide and water?

(A) 4.5 mol (B) 6.0 mol
(C) 7.5 mol (D) 8.0 mol

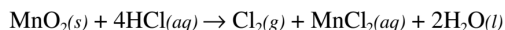
151. A 0.200 g sample of benzoic acid, $\text{C}_6\text{H}_5\text{COOH}$, is titrated with a 0.120 M $\text{Ba}(\text{OH})_2$ solution. What volume of the $\text{Ba}(\text{OH})_2$ solution is required to reach the equivalence point?

(A) 6.82 mL (B) 13.6 mL
(C) 17.6 mL (D) 35.2 mL

Substance	Molar Mass
$\text{C}_6\text{H}_5\text{COOH}$	$122.1 \text{ g} \cdot \text{mol}^{-1}$

Stoichiometry/Solutions

152. Chlorine can be prepared by reacting HCl with MnO_2 . The reaction is represented by this equation.



Assuming the reaction goes to completion what mass of concentrated HCl solution (36.0% HCl by mass) is needed to produce 2.50 g of Cl_2 ?

- (A) 5.15 g (B) 14.3 g
(C) 19.4 g (D) 26.4 g

153. What is the Na^+ ion concentration in the solution formed by mixing 20. mL of 0.10 M Na_2SO_4 solution with 50. mL of 0.30 M Na_3PO_4 solution?

- (A) 0.15 M (B) 0.24 M
(C) 0.48 M (D) 0.70 M

154. A solution prepared by dissolving a 2.50 g sample of an unknown compound dissolved in 34.0 g of benzene, C_6H_6 , boils 1.38°C higher than pure benzene. Which expression gives the molar mass of the unknown compound?

Compound	K_b
C_6H_6	$2.53^\circ\text{C}\cdot\text{m}^{-1}$

- (A) $2.53 \times \frac{2.50}{1.38}$
(B) $1.38 \times \frac{34.0}{2.53} \times 2.50$
(C) $2.50 \times 10^3 \times \frac{2.53}{34.0} \times \frac{1}{1.38}$
(D) $2.50 \times 10^3 \times \frac{1.38}{34.0} \times 2.53$

155. A 20.00 mL sample of a $\text{Ba}(\text{OH})_2$ solution is titrated with 0.245 M HCl. If 27.15 mL of HCl is required, what is the molarity of the $\text{Ba}(\text{OH})_2$ solution?

- (A) 0.166 M (B) 0.180 M
(C) 0.333 M (D) 0.666 M

156. 0.250 g of an element, M, reacts with excess fluorine to produce 0.547 g of the hexafluoride, MF_6 . What is the element?

- (A) Cr (B) Mo (C) S (D) Te

157. How many moles of Na^+ ions are in 20 mL of 0.40 M Na_3PO_4 ?

- (A) 0.0080 (B) 0.024
(C) 0.050 (D) 0.20

158. What is the mass percent of oxygen in $\text{Al}_2(\text{SO}_4)_3 \cdot 18\text{H}_2\text{O}$?

Molar Mass, M
$\text{Al}_2(\text{SO}_4)_3 \cdot 18\text{H}_2\text{O}$ 666.43 g·mol

- (A) 9.60 (B) 28.8 (C) 43.2 (D) 72.0

159. What is the number of O_2 molecules in the 2.5 g of O_2 inhaled by the average person in one minute?

- (A) 1.9×10^{22} (B) 3.8×10^{22}
(C) 4.7×10^{22} (D) 9.4×10^{22}